

Past-Unbounded Cosmology

An Exact Dust-Vacuum Return Condition

Abhijit Singh

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Abstract

A past-unbounded causal history is separated from the finite age of a particular cosmological branch. In a flat dust-plus-vacuum model, acceleration onset and matter-vacuum equality are distinct. Applying a declared complement return to the acceleration boundary gives an exact present vacuum fraction of two thirds. Without fitting within the branch, this condition forces $z_{\text{acc}} = 0.587401052$, $z_{\text{eq}} = 0.259921050$, $H_0 t_0 = 0.9358813101$, $\Delta N = 0.4620981204$, $q_0 = -1/2$, and $j_0 = 1$. A nonsingular recurrent solution demonstrates mathematical consistency of a past-unbounded class. The paper also proves the absolute-scale obstruction, separates cold-matter and slow-roll requirements, and states a full-path falsification protocol. Black-hole discussion is confined to accessibility and known thermodynamic targets.

Keywords: cosmology; dark energy; cosmic age; acceleration; recurrence; horizons

Scope and status: The numerical values are exact conditional consequences of the displayed model and returned-present postulate. They are not an observational fit. Absolute scale, perturbations, and a full likelihood remain open.

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1 Past-unbounded histories and typed age

A causal history is past-unbounded when

$$\forall e \exists e' \quad e' \prec e. \quad (1)$$

This condition is compatible with finite ages for particular thermodynamic or geometric phases. A total age function cannot be both invariant under arbitrary origin shift and increase by one under every update. A past-unbounded model therefore has no unique origin-independent total age, while phase ages remain well defined relative to physical transitions.

2 Flat dust-vacuum branch

Consider a spatially flat Friedmann model with pressureless matter and a constant vacuum term:

$$H(a)^2 = H_0^2 \left(\Omega_{m,0} a^{-3} + \Omega_{\Lambda,0} \right), \quad \Omega_{m,0} + \Omega_{\Lambda,0} = 1. \quad (2)$$

Let $p = \rho_{\Lambda}/(\rho_m + \rho_{\Lambda})$. In e-fold time $N = \log a$,

$$p' = 3p(1 - p). \quad (3)$$

Acceleration begins at $p = 1/3$. The declared complement return maps this boundary to

$$\boxed{p_{\text{ret}} = 2/3.} \quad (4)$$

Identifying the present epoch with the returned event is the model's physical postulate. The consequences below are exact after that postulate.

3 Exact dimensionless predictions

Matter-vacuum equality satisfies $\rho_m = \rho_{\Lambda}$, while acceleration onset satisfies $\rho_m = 2\rho_{\Lambda}$. With $(\Omega_{m,0}, \Omega_{\Lambda,0}) = (1/3, 2/3)$,

$$z_{\text{eq}} = 2^{1/3} - 1 = 0.259921050, \quad (5)$$

$$z_{\text{acc}} = 2^{2/3} - 1 = 0.587401052. \quad (6)$$

The age integral is

$$H_0 t_0 = \int_0^1 \frac{da}{a \sqrt{\Omega_{m,0} a^{-3} + \Omega_{\Lambda,0}}} = \frac{2}{3 \sqrt{\Omega_{\Lambda,0}}} \operatorname{arsinh} \sqrt{\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}}. \quad (7)$$

Therefore

$$\boxed{H_0 t_0 = \sqrt{\frac{2}{3}} \operatorname{arsinh} \sqrt{2} = 0.9358813101.} \quad (8)$$

The logarithmic expansion interval associated with the returned fraction is

$$\boxed{\Delta N = \frac{2}{3} \ln 2 = 0.4620981204.} \quad (9)$$

The present deceleration and jerk are

$$q_0 = \frac{1}{2} \Omega_{m,0} - \Omega_{\Lambda,0} = -\frac{1}{2}, \quad j_0 = 1. \quad (10)$$

The fraction of branch age elapsed since acceleration began is 0.4255192360.

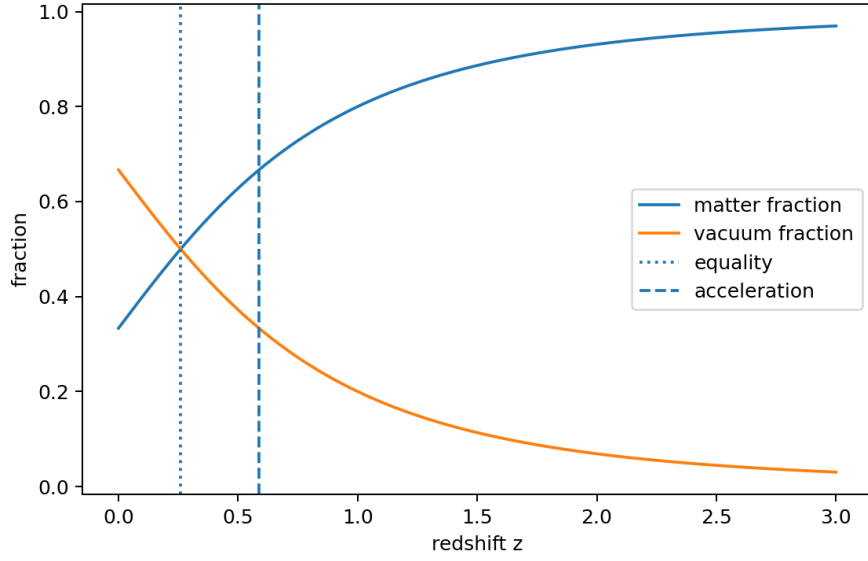


Figure 1: Exact matter and vacuum fractions in the returned dust-vacuum branch.

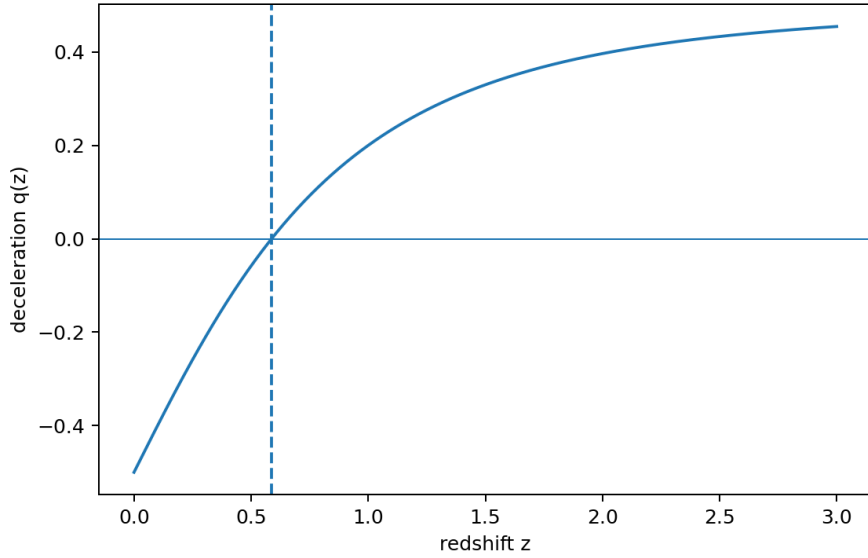


Figure 2: Deceleration parameter; the zero crossing is the exact acceleration redshift.

4 Comparison with one external parameter estimate

Using the Planck 2018 flat- Λ CDM estimate $\Omega_{\Lambda,0} \approx 0.6847 \pm 0.0073$, the value $2/3$ differs by approximately 2.47 quoted standard deviations. This comparison is model dependent and not a full likelihood analysis. It makes the returned-present postulate sharp and already constrained rather than adjustable.

5 Scale obstruction

All results above are dimensionless. The dilation

$$H \mapsto \mu H, \quad t \mapsto t/\mu \quad (11)$$

leaves the dimensionless history invariant. No absolute H_0 , age in years, vacuum energy density, sound horizon, or physical parent scale follows without one additional dimensionful input generated by the model.

A later controlled transmutation map gives an exact conditional relation between a symmetry scale, coupling data, and an absolute age. The physical vacuum formation and its parameters are not yet derived, so the map is retained as a conditional calibration route rather than a numerical prediction.

6 Past-unbounded recurrent example

The equation

$$\ddot{\chi} + \omega^2(\chi - \chi_*) = 0 \quad (12)$$

has $\chi = \chi_* + A \cos(\omega t + \phi)$. Setting $J_g = e^\chi$ gives a positive scale factor

$$a(t) = \exp \left[\frac{\chi_* + A \cos(\omega t + \phi)}{d_s} \right], \quad (13)$$

with alternating expansion and contraction. This proves mathematical consistency of a nonsingular past-unbounded class; it does not identify the observed universe.

7 Dark sectors

A cold dark-matter component must have negligible pressure, cluster gravitationally, and approximately scale as a^{-3} . A coherently oscillating massive scalar can reproduce this background and clustering only when its mass is large relative to the Hubble scale on the relevant epochs.

A late dark-energy component with $w \simeq -1$ requires kinetic energy much smaller than potential energy and an effective mass of order H_0 or below if rolling today. One homogeneous quadratic mode cannot simultaneously satisfy both present-day cold-matter clustering and slow-roll dark energy. A unified origin would require distinct modes, phases, condensates, or a constant vacuum term plus an oscillating component.

8 Horizons and black holes

An event horizon is a boundary of future accessibility,

$$\mathcal{H} = \partial J^-(I^+). \quad (14)$$

Exterior inaccessibility is not identical to global information destruction. A complete microscopic theory must recover

$$S_{\text{BH}} = \frac{k_B A}{4\ell_P^2}, \quad (15)$$

Hawking dynamics, interior continuation, and unitary accounting. The present framework supplies an accessibility interpretation and a two-exponent distinction between residual instability and inaccessible provenance; it does not derive black-hole microstates or evaporation.

9 Full-path falsification

The branch must be tested without retuning the returned fraction. The linked functions

$$H(z), \quad q(z), \quad D_A(z), \quad D_L(z), \quad t(z), \quad f\sigma_8(z) \quad (16)$$

should be evaluated against a frozen external likelihood. Failure of any linked observable rejects the branch as a physical account while leaving the conditional algebra intact.

10 Status

The $2/3$ return and all displayed dimensionless numbers are exact inside the flat dust-vacuum and present-equals-return assumptions. The present-equals-return postulate, absolute scale, perturbation spectrum, sound horizon, and full likelihood remain empirical or open.

11 Typed age and the absence of a universal origin

A causal history is past-unbounded when every event has an earlier event. Such a history can contain finite-duration phases without a first event. An origin-independent total age A cannot both satisfy translation invariance $A(n+k) = A(n)$ for arbitrary origin shifts and increment law $A(n+1) = A(n) + 1$. Hence a past-unbounded model has no unique total age while a hot, smooth, or observational branch can have a finite age measured from a physical transition.

12 Exact FLRW derivation

For flat dust plus a constant vacuum component,

$$E(a)^2 = \frac{H(a)^2}{H_0^2} = \Omega_{m,0}a^{-3} + \Omega_{\Lambda,0}, \quad \Omega_{m,0} + \Omega_{\Lambda,0} = 1. \quad (17)$$

Under the returned-state condition

$$\Omega_{\Lambda,0} = \frac{2}{3}, \quad \Omega_{m,0} = \frac{1}{3}, \quad (18)$$

the matter-vacuum equality condition gives

$$1 + z_{\text{eq}} = 2^{1/3}, \quad (19)$$

and the acceleration condition $\rho_m = 2\rho_\Lambda$ gives

$$1 + z_{\text{acc}} = 2^{2/3}. \quad (20)$$

The age integral is

$$H_0 t_0 = \int_0^1 \frac{da}{aE(a)} = \frac{2}{3\sqrt{\Omega_{\Lambda,0}}} \operatorname{arsinh} \sqrt{\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}}, \quad (21)$$

so

$$H_0 t_0 = \sqrt{\frac{2}{3}} \operatorname{arsinh} \sqrt{2} = 0.9358813101. \quad (22)$$

The exact branch register is

quantity	value
$\Omega_{\Lambda,0}$	$2/3$
z_{acc}	0.587401052
z_{eq}	0.259921050
$H_0 t_0$	0.9358813101
q_0	$-1/2$
j_0	1
ΔN	$(2/3) \ln 2 = 0.4620981204$

These are forced by the branch assumptions rather than fitted independently.

13 Past-unbounded recurrent construction

The equation

$$\ddot{\chi} + \omega^2(\chi - \chi_*) = 0 \quad (23)$$

has $\chi = \chi_* + A \cos(\omega t + \phi)$. Setting $J_g = e^\chi$ and $a = J_g^{1/d_s}$ gives a positive nonsingular scale factor for all t , with alternating expansion and contraction. This proves consistency of one past-unbounded cosmological class; it does not identify the observed universe with that class.

14 Finite-capacity fate theorem

If accessible geometric capacity is finite but $H(t) \geq h > 0$ for all sufficiently late time, then

$$J_g(t) \geq J_g(t_0)e^{d_s h(t-t_0)}, \quad (24)$$

which eventually exceeds every finite bound. Thus finite capacity requires $H \rightarrow 0$, a later sign change, or continuing access to new capacity.

15 Dark sectors and horizon scope

A cold dark-matter candidate must cluster and approximately scale as a^{-3} ; the framework supplies a possible structural role for stable neutral recurrent excitations but no mass, abundance, or cross section. A vacuum component has $p = -\rho$; the two-thirds branch fixes a dimensionless present fraction but does not derive the observed microscopic vacuum scale.

An event horizon is a boundary of future accessibility. Exterior inaccessibility is not identical to global destruction of information. A microscopic theory must recover the area law, its coefficient, Hawking flux, and unitary accounting. The present causal description identifies the boundary problem but does not derive those microphysics.

16 Full observational test

Freeze the 2/3 postulate before comparison. Evaluate the linked functions $H(z)$, $q(z)$, distance measures, age, and growth under a current external likelihood without retuning the return condition. Model comparison must include covariance, priors, and alternatives. A failure of any linked observable rejects the physical branch while leaving the conditional derivation intact.

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